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Total No. of Questions : 13

B.Pharmacy (Sem.–3)

PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 75106

Date of Examination:18-12-2024

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Define briefly:**

   a) What is meant by intrinsic solubility?

   b) State Raoult's Law with its applications.

   c) What is meant by interfacial tension?

   d) What are chelates? Give two examples.

   e) What is absolute humidity?

   f) What is glass transition temperature?

   g) What are polar and semi-polar solvents? Give two examples each.

   h) Define HLB value and its scale.

   i) Differentiate between physical adsorption and chemisorption.

   j) What is spreading coefficient?

SECTION-B

2. Discuss the pH titration method for determination of stoichiometric ratio in a complex.

3. What is an 'ideal solution'? Differentiate between complete miscibility and partial miscibility phenomena taking suitable examples.

4. Differentiate between surface tension and interfacial tension. With the help of suitable equations explain spreading of one liquid on another liquid.

SECTION-C

5. Discuss the calculation of buffer capacity and its significance.

6. A 100 ml solution of ephedrine sulphate is to be made isotonic. How much dextrose should be added for this purpose? (Given: 'E' value of ephedrine sulphate is 0.23 and of dextrose is 0.16).

7. What are amorphous and crystalline solids? Give examples of polymorphism and highlight the advantages and disadvantages of polymorphic behavior of solids.

8. Explain self-association phenomena.

9. Classify surfactants and write a note on detergency.

10. Write a note on dissociation constant and its significance.

11. What is glassy state of a solid? Explain the properties of such a solid and its significance in dosage form performance.

12. Differentiate between solvation and association with examples.

13. Briefly discuss the factors influencing the solubility of drugs.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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